Conditional flow matching vs. diffusion models

A systematic comparison on CIFAR-10

Ali Dor Elora Drouilhet

March 23, 2026

CentraleSupélec/MVA — Advanced Deep Learning course

Outline

Motivation

Background

Our implementation

2D toy experiments

CIFAR-10 results

ODE inversion & image editing

Analysis

Conclusion

Motivation

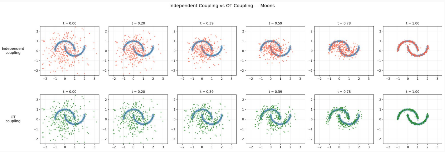
Why flow matching?

The problem with diffusion models:

- DDPM needs many steps to generate one image
- DDIM reduces steps but quality saturates at a certain point
- Curved denoising paths are hard to integrate

Flow matching promise:

- Learn **straight** transport paths from noise to data
- ODE-based: use any solver, any # steps
- OT coupling $\Rightarrow$ nearly linear trajectories
- **Same quality, fewer steps**



Independent vs. OT coupling on 2D data

Background

DDPM & DDIM

DDPM — Denoising Diffusion Probabilistic Model:

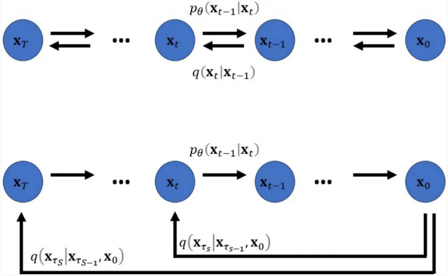
- Gradually adds Gaussian noise to data over $T=1000$ steps
- Trains a network ϵ_θ to predict the added noise:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2\right]$$

- Generates by iteratively denoising $\mathbf{x}_T \to \mathbf{x}_0$
- Slow: requires all 1000 steps

DDIM — deterministic shortcut:

- Reuses the same trained DDPM model
- Skips timesteps: 10–100 steps instead of 1000
- But: approximation errors accumulate, quality saturates



The forward and reverse processes of DDPM (top) and DDIM (bottom)

Flow matching: a different paradigm

Key idea: learn a velocity field, not noise

OT interpolation path:

$$\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1$$

$$\mathbf{u}_t = \mathbf{x}_1 - \mathbf{x}_0 \quad \text{(constant velocity!)}$$

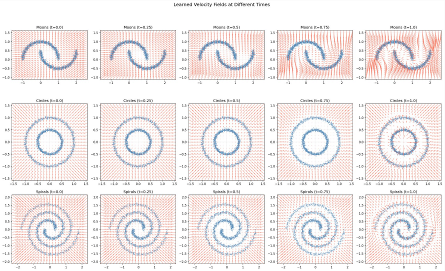
Training (predict velocity):

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1}\left[\|v_\theta(\mathbf{x}_t, t) - \mathbf{u}_t\|^2\right]$$

Sampling: solve ODE from $t=0 \to 1$

$$\frac{d\mathbf{x}}{dt} = v_\theta(\mathbf{x}_t, t)$$

- **Straight paths** $\Rightarrow$ easy to integrate
- Any ODE solver, any number of steps



Learned velocity fields at different t

ODE solvers for flow matching

Table: ODE solvers compared

Solver	Order	NFE/step	Best for
Euler	1st	1	Simplest baseline
Midpoint	2nd	2	Low NFE (10–50)
RK4	4th	4	High NFE (100+)

Key insight: at fixed NFE budget, higher-order solvers take fewer but more accurate steps.

Our implementation

Architecture

UNet (DDPM/DDIM & unconditional CFM):

- 128 base channels
- Channel mult: $[1, 2, 2, 2]$
- 4 attention heads, GroupNorm-32
- Sinusoidal time embeddings
- Adam, $\text{lr} = 2 \times 10^{-4}$
- 500 epochs, EMA 0.9999

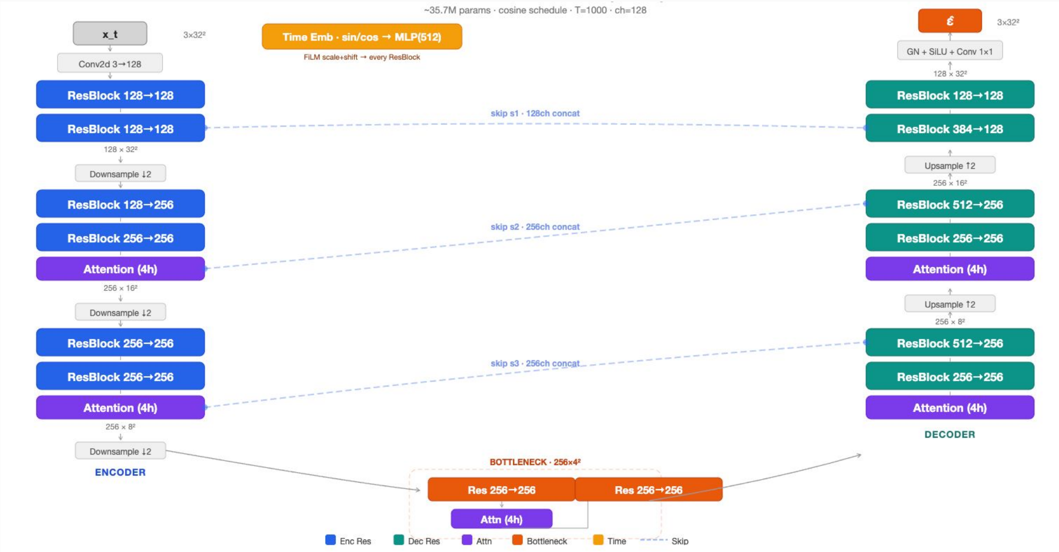
Evaluation protocol:

- 10,000 generated samples
- FID and Inception Score
- NFE sweep: 10, 20, 50, 100
- Google Colab (T4 / A100)

SimpleFlowUNet (class-conditional CFM):

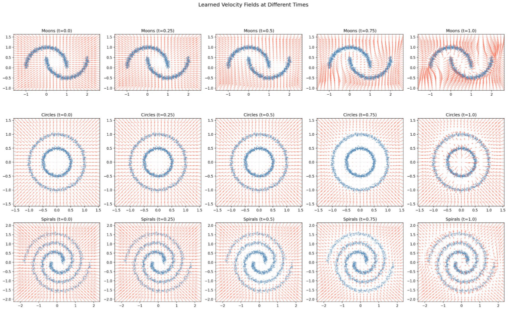
- 128 base channels
- Channel mult: $[1, 2, 4]$
- Single-head attention, GroupNorm-8
- Learned class embeddings (10 classes)
- 10% class dropout $\Rightarrow$ CFG
- AdamW, $\text{lr} = 2 \times 10^{-4}$
- 210 epochs, EMA 0.999

Architecture

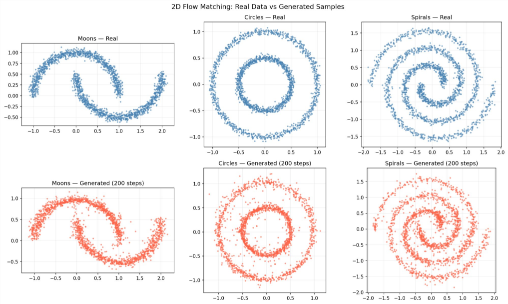


2D toy experiments

Learning to transport: 2D visualization



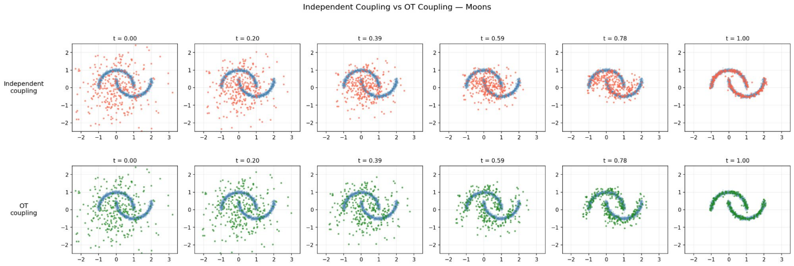
Velocity fields at $t = 0.0, 0.25, 0.5, 0.75, 1.0$



Generated 2D samples vs. target distribution

The model learns time-varying velocity fields that smoothly transport Gaussian noise to complex multi-modal distributions.

Why optimal transport matters



Independent coupling

Paths cross $\Rightarrow$ curved trajectories

Harder for ODE solvers

OT coupling

Non-crossing, straight paths

Easy to integrate accurately

CIFAR-10 results

Main results: FID vs NFE

Table: Full comparison on CIFAR-10

Method	Solver	NFE	FID ↓
DDPM	—	1000	13.06
DDIM	—	10	22.18
DDIM	—	20	18.95
DDIM	—	50	17.96
DDIM	—	100	18.43 ↑
OT-CFM	Midpoint	10	17.53
OT-CFM	Midpoint	20	14.80
OT-CFM	Midpoint	50	12.02
OT-CFM	Midpoint	100	11.21
OT-CFM	RK4	100	**10.65**
Cond. CFM	Midpoint	50	13.17
Cond. CFM	RK4	100	12.19

Key findings:

Best FID: 10.65
(OT-CFM, RK4, 100 NFE)

vs. DDPM: 13.06 at 1000 NFE
$\Rightarrow$ **20**$\times$ **speedup**

DDIM degrades past 50 NFE
(18.43 at 100 vs. 17.96 at 50)

OT-CFM at **20 NFE** (14.80)
already beats best DDIM (17.96)

Best results summary

Table: Best FID per model family

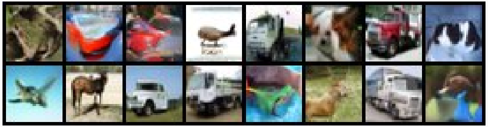
Model	Best FID	NFE	Solver	ms/img
DDPM	13.06	1000	—	205.0
DDIM	17.96	50	—	10.5
OT-CFM	**10.65**	**100**	**RK4**	**20.7**
Cond. CFM	12.19	100	RK4	24.3

10× fewer steps, better quality

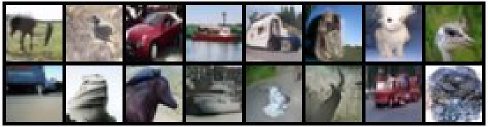
DDPM
1000 NFE

OT-CFM
100 NFE

Generation quality: samples

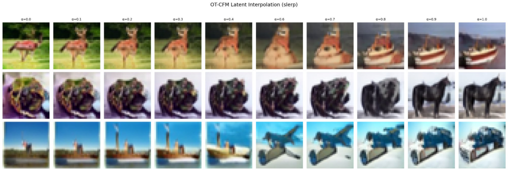


OT-CFM unconditional samples (epoch 500)

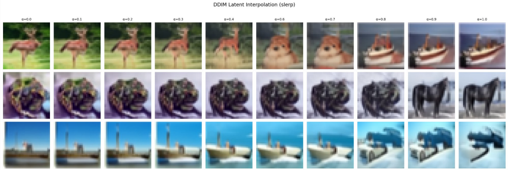


DDPM samples (epoch 500)

Latent interpolation



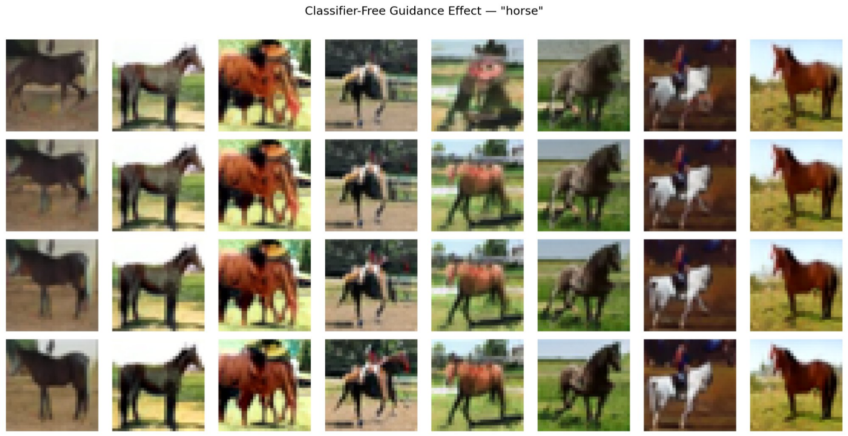
OT-CFM interpolation



DDIM interpolation

Both models support smooth latent interpolation, but CFM transitions are notably smoother due to the straight OT paths.

Class-conditional generation with CFG



Effect of classifier-free guidance on class-conditional generation

$$v = v_{\text{uncond}} + s\left(v_{\text{cond}} - v_{\text{uncond}}\right)$$

ODE inversion & image editing

ODE inversion: the key idea

CFM's ODE is **deterministic and invertible**:

Forward (generation):

$$z_0 \xrightarrow{t=0\to1} x_1$$

Backward (inversion):

$$x_1 \xrightarrow{t=1\to0} z_0$$

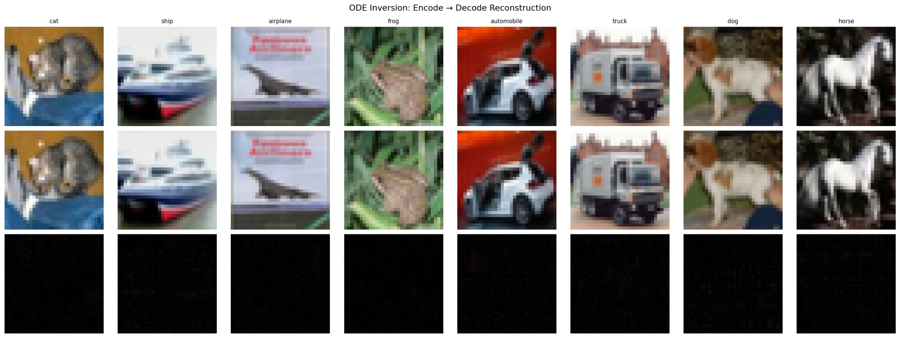
Every real image has a unique "address" in noise space.

This enables:

- Encode any real image to noise
- Edit by manipulating the noise
- Decode back to get edited image

Real images dissolving into noise (t=1→0)

Reconstruction: encode → decode

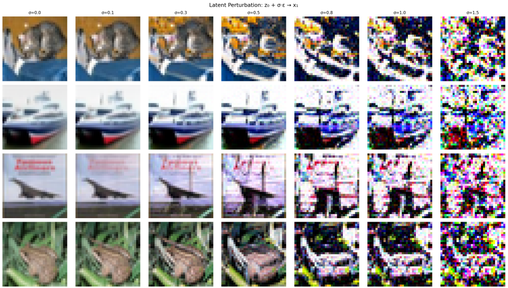


- Top: original CIFAR-10 images (one per class)
- Middle: reconstructed via backward ODE → forward ODE
- Bottom: pixel difference (5× amplified) — **near-zero**

The ODE inversion is **near-perfect**, confirming the model learned an accurate invertible flow.

Latent perturbations & attribute editing

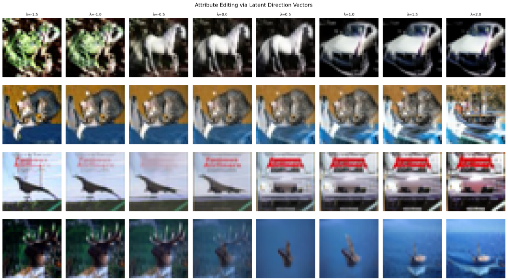
Latent perturbations



Perturb $z_0 + \sigma\epsilon$, then decode.
σ=0: original. σ=0.3: subtle change.
σ=1.5: content destroyed.

Latent space is smooth and locally stable.

Attribute editing



$$z_0^{\text{edit}} = z_0 + \lambda(\bar{z}_0^{\text{tgt}} - \bar{z}_0^{\text{src}})$$

λ=0: original. $\lambda>0$: toward target.
e.g. horse$\to$car, cat$\to$dog.

Subtle but directional at 32×32.

Real image interpolation

Encode two real images $\to z_0^A,\ z_0^B$, then decode blends: $z_0^\alpha = (1-\alpha)\, z_0^A + \alpha\, z_0^B$

Analysis

Why does flow matching win?

Diffusion (DDPM/DDIM)

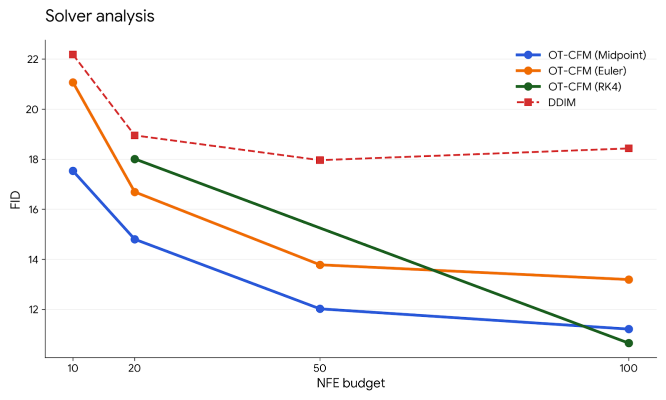
- Learns noise prediction ϵ_θ
- Curved denoising paths
- DDIM: deterministic but approximate
- Quality saturates at $\sim$50 steps
- FID degrades past 50 NFE

Flow Matching (OT-CFM)

- Learns velocity field v_θ
- Straight OT paths
- Exact ODE formulation
- Quality keeps improving with NFE
- Best at every NFE budget

Root cause: OT-CFM's straight paths are easy for ODE solvers.
DDIM inherits errors from stochastic training $\Rightarrow$ error accumulation.

Solver comparison & DDIM saturation



Solver choice:

- **Midpoint**: best at low NFE (10–50)
- **RK4**: best absolute FID at 100 NFE
- **Euler**: simplest, worst per-NFE

DDIM saturation:

- FID worsens from 50→100 NFE
- Stochastic training + deterministic sampling ⇒ error accumulation
- CFM: no mismatch, FID keeps improving

Recommendation: Midpoint for fast, RK4 for best quality.

Conclusion

Key takeaways

1. **Flow matching** $>$ **diffusion** at every NFE budget on CIFAR-10
2. **20**$\times$ **speedup**: OT-CFM matches DDPM quality in $\sim$50 steps vs. 1000
3. **Best FID: 10.65** (OT-CFM $+$ RK4 at 100 NFE)
4. **Midpoint solver** recommended for low-NFE regimes
5. **CFG improves class fidelity** while trading off diversity
6. **ODE inversion** enables image editing, a unique advantage of flow matching

Limitations & future work

Limitations:

- Single dataset: CIFAR-10 at 32×32
- No confidence intervals (single runs)
- Conditional model undertrained (210 vs. 500 epochs)

Future directions:

- Scale to higher resolutions (64×64, 256×256)
- Explore σ_{min} and advanced OT solvers
- Adaptive step-size ODE solvers
- Combine with latent diffusion / VAE framework

What each of us did

Ali Dor

- DDPM / DDIM implementation
- ODE inversion & image editing
- Visualizations:
  - 2D toy experiments
  - Generation trajectories
  - Latent interpolations
- Systematic NFE/solver evaluation
- Results analysis

Elora Drouilhet

- OT-CFM implementation
- Class-conditional CFM
  - Class embeddings
  - Classifier-free guidance
- Model training & evaluation
- CFG scale analysis
- Results analysis

Thank you!

Questions?

Ali Dor & Elora Drouilhet
CentraleSupélec

References

Y. Lipman et al.
Flow matching for generative modeling.
In *ICLR*, 2023.

A. Tong et al.
Improving and generalizing flow-based generative models with minibatch optimal transport.
TMLR, 2024.

J. Ho et al.
Denoising diffusion probabilistic models.
In *NeurIPS*, 2020.

J. Song et al.
Denoising diffusion implicit models.
In *ICLR*, 2021.